



DS-SERIES

PROFESSIONAL OUTDOOR LOUDSPEAKER

USER MANUAL

DS-5 · DS-8 · DS-12 · DS-15



IP54 WEATHERPROOF RATED

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TABLE OF CONTENTS

1.	Introduction & Product Overview
2.	Safety Instructions
3.	DS-5 Outdoor IP54 Loudspeaker
4.	DS-8 Outdoor IP54 Loudspeaker
5.	DS-12 Outdoor IP54 Loudspeaker
6.	DS-15 Outdoor IP54 Loudspeaker
7.	Full Specifications Comparison
8.	Installation Guidelines
9.	Warranty Information
10.	Contact & Support

1. Introduction & Product Overview

The Louis Martin DS-Series represents a complete line of professional-grade outdoor IP54-rated loudspeakers, engineered to deliver outstanding audio performance in demanding environments. Ranging from the compact DS-5 to the powerful DS-15, each model features a coaxial two-way driver configuration with a passive crossover optimised for linearity, ensuring exceptional vocal clarity, high fidelity, and minimal distortion across all operating conditions.

All four models share the same IP54 weatherproof ABS plastic enclosure philosophy and wide-dispersion horn design, making them equally at home in background music installations, sports complexes, educational facilities, theme parks, and high-power outdoor events.

Key Series Features

- IP54-rated weatherproof ABS plastic cabinet
- Two-way coaxial driver with passive crossover optimised for linearity
- Nominal impedance: 8 Ω (typical) across all models
- Multiple mounting configurations with included bracket hardware
- Available in Black and White colour options (DS-12 / DS-15)
- Two-year manufacturer's warranty on workmanship and materials

Intended Applications

- Outdoor & indoor background music — cafés, terraces, hospitality
- Gyms, fitness centres, and classrooms (DS-5, DS-8)
- Stadiums, sport complexes, museums, and theme parks (DS-12, DS-15)
- High-power outdoor public address systems

2. Safety Instructions

Flying & Suspension

When flying speakers, appropriate certified hardware and rigging techniques must be employed to safely suspend the enclosures. All hardware and flying apparatus must be regularly inspected for any visual or mechanical failure before each use.

Audience Distance

Maintain adequate distance between loudspeakers and the audience at all times. Consult your local Health & Safety authority for guidance on safe sound-pressure levels in public spaces.

Controller Programming

The Louis Martin digital controller must only be programmed by an authorised technician. Operating the speaker system without the correct controller and programme will void the warranty.

Environmental Exposure

Although rated IP54, avoid unnecessary prolonged exposure to standing water, high-pressure jets, or submerged conditions. Protect all connected electronics from humidity and rain.

Electrical Safety

Ensure all amplifier and wiring ratings are compatible with the loudspeaker's impedance and power specifications. Never exceed the stated peak power handling.

Children & Unauthorised Access

Keep children and untrained personnel away from suspended loudspeaker systems. Never allow unauthorised persons to alter the mounting hardware or cabling.

3. DS-5 Outdoor IP54 Loudspeaker

The DS-5 is a compact yet powerful outdoor loudspeaker built around a 5.25" (133 mm) coaxial driver paired with a 1.34" (34 mm) high-frequency driver. Its wide 115° × 115° dispersion horn ensures even coverage across both horizontal and vertical planes, making it the ideal solution for background music, café terraces, gym areas, and classrooms. The IP54-rated ABS plastic cabinet allows permanent outdoor installation without the risk of moisture damage.



System Applications

- Outdoor & indoor background music
- Terrace cafés and hospitality venues
- Gym and fitness centres
- Classrooms and educational facilities

Technical Specifications

Type	Two-way Coaxial full range plastic IP54 loudspeaker
Frequency Response	65 Hz – 19 kHz (–10 dB)
LF Driver	1 × 5.25" (133 mm) Coaxial Driver
HF Driver	1 × 1.34" (34 mm) HF Driver
RMS Power	100 W
Program Power	200 W
Peak Power	400 W
Sensitivity (1W/1m)	93 dB
Max SPL Continuous (1m)	115 dB

Max SPL Peak (1m)	118 dB
Nominal Impedance	8 Ω (typical)
Crossover Mode	Passive, optimised for linearity
Dispersion (H $\times$ V)	115° (H) $\times$ 115° (V)
Cabinet Material	ABS Plastic — IP54 rated
Dimensions (H $\times$ W $\times$ D)	328 $\times$ 192 $\times$ 160 mm
Weight	5 kg / 11.02 lbs

Cabinet Views & Dimensions



Front View



Rear View



Side View



Top View

Cabinet Dimensions: 328 $\times$ 192 $\times$ 160 mm · Weight: 5 kg / 11.02 lbs

4. DS-8 Outdoor IP54 Loudspeaker

The DS-8 steps up to an 8" (203 mm) coaxial driver, delivering extended low-frequency response down to 60 Hz and a 50 W increase in RMS handling over the DS-5. With the same 115° × 115° wide-dispersion horn, it covers larger outdoor spaces while maintaining tonal consistency. The DS-8 is an excellent choice for medium-area hospitality and educational installations that demand greater headroom and deeper bass.



System Applications

- Outdoor applications requiring greater headroom
- Background music in medium-to-large venues
- Terrace cafés, gym and fitness centres
- Classrooms and lecture halls

Technical Specifications

Type	Two-way Coaxial full range plastic IP54 loudspeaker
Frequency Response	60 Hz – 19 kHz (–10 dB)
LF Driver	1 × 8" (203 mm) Coaxial Driver
HF Driver	1 × 1.34" (34 mm) HF Driver
RMS Power	150 W
Program Power	300 W
Peak Power	600 W
Sensitivity (1W/1m)	95 dB
Max SPL Continuous (1m)	118 dB

Max SPL Peak (1m)	121 dB
Nominal Impedance	8 Ω (typical)
Crossover Mode	Passive, optimised for linearity
Dispersion (H $\times$ V)	115° (H) $\times$ 115° (V)
Cabinet Material	ABS Plastic — IP54 rated
Dimensions (H $\times$ W $\times$ D)	470 $\times$ 230 $\times$ 260 mm
Weight	8 kg / 17.6 lbs

Cabinet Views & Dimensions



Front View



Rear View



Side View



Top View

Cabinet Dimensions: 470 $\times$ 230 $\times$ 260 mm · Weight: 8 kg / 17.6 lbs

5. DS-12 Outdoor IP54 Loudspeaker

The DS-12 is a high-power coaxial loudspeaker engineered for large-scale outdoor installations. Its 12" (304 mm) coaxial driver with upgraded 1.45" (37 mm) HF driver produces up to 128 dB SPL continuously at 1 m. The narrower 90° × 60° horn pattern provides controlled directivity, reducing spill in demanding acoustic environments such as stadiums, sport complexes, and theme parks. Available in Black or White.



System Applications

- Theme parks and large outdoor events
- Stadiums and sport complexes
- Museums and exhibition halls
- High-power outdoor public address

Technical Specifications

Type	Two-way Coaxial full range plastic IP54 loudspeaker
Frequency Response	70 Hz – 19 kHz (–10 dB)
LF Driver	1 × 12" (304 mm) Coaxial Driver
HF Driver	1 × 1.45" (37 mm) HF Driver
RMS Power	400 W
Program Power	800 W
Peak Power	1600 W
Sensitivity (1W/1m)	99 dB
Max SPL Continuous (1m)	128 dB

Max SPL Peak (1m)	131 dB
Nominal Impedance	8 Ω (typical)
Crossover Mode	Passive, optimised for linearity
Dispersion (H × V)	90° (H) × 60° (V)
Cabinet Material	ABS Plastic — IP54 rated
Dimensions (H × W × D)	425 × 415 × 435 mm
Weight	17 kg / 37.4 lbs

Cabinet Views & Dimensions



Cabinet Dimensions: 425 × 415 × 435 mm · Weight: 17 kg / 37.4 lbs

6. DS-15 Outdoor IP54 Loudspeaker

At the apex of the DS-Series, the DS-15 houses a massive 15" (381 mm) coaxial driver and an enlarged 1.75" (44 mm) HF driver, delivering 500 W RMS and peaks of 135 dB SPL — the highest in the series. The 90° × 60° controlled-dispersion horn, combined with the IP54 enclosure, makes the DS-15 the definitive choice for stadiums, sport complexes, and any application that demands maximum output with weatherproof reliability. Available in Black or White.



System Applications

- Stadiums and major sport complexes
- Theme parks and large open-air venues
- Museums and high-ceiling exhibition halls
- Maximum-output outdoor public address

Technical Specifications

Type	Two-way Coaxial full range plastic IP54 loudspeaker
Frequency Response	65 Hz – 19 kHz (–10 dB)
LF Driver	1 × 15" (381 mm) Coaxial Driver
HF Driver	1 × 1.75" (44 mm) HF Driver
RMS Power	500 W
Program Power	1000 W
Peak Power	2000 W
Sensitivity (1W/1m)	102 dB
Max SPL Continuous (1m)	132 dB

Max SPL Peak (1m)	135 dB
Nominal Impedance	8 Ω (typical)
Crossover Mode	Passive, optimised for linearity
Dispersion (H $\times$ V)	90° (H) $\times$ 60° (V)
Cabinet Material	ABS Plastic — IP54 rated
Dimensions (H $\times$ W $\times$ D)	490 $\times$ 490 $\times$ 520 mm
Weight	25 kg / 55.1 lbs

Cabinet Views & Dimensions



Front View



Rear View



Side View

Cabinet Dimensions: 490 $\times$ 490 $\times$ 520 mm · Weight: 25 kg / 55.1 lbs

7. Full Specifications Comparison

The table below provides a side-by-side overview of all four DS-Series models to assist with system design and model selection.

Specification	DS-5	DS-8	DS-12	DS-15
Driver (LF)	5.25" (133mm)	8" (203mm)	12" (304mm)	15" (381mm)
Driver (HF)	1.34" (34mm)	1.34" (34mm)	1.45" (37mm)	1.75" (44mm)
Freq. Response	65–19 kHz	60–19 kHz	70–19 kHz	65–19 kHz
RMS Power	100 W	150 W	400 W	500 W
Program Power	200 W	300 W	800 W	1000 W
Peak Power	400 W	600 W	1600 W	2000 W
Sensitivity	93 dB	95 dB	99 dB	102 dB
Max SPL (Cont.)	115 dB	118 dB	128 dB	132 dB
Max SPL (Peak)	118 dB	121 dB	131 dB	135 dB
Impedance	8 Ω	8 Ω	8 Ω	8 Ω
Dispersion H×V	115°×115°	115°×115°	90°×60°	90°×60°
Dimensions (H×W×D)	328×192×160mm	470×230×260mm	425×415×435mm	490×490×520mm
Weight	5 kg / 11 lbs	8 kg / 17.6 lbs	17 kg / 37.4 lbs	25 kg / 55.1 lbs
IP Rating	IP54	IP54	IP54	IP54

8. Installation Guidelines

8.1 Mounting & Rigging

Each DS-Series loudspeaker ships with a multi-angle bracket that supports wall, ceiling, and pole-mount configurations. For permanent flying installations, use only certified M10 or larger stainless steel hardware rated to a minimum safety factor of 10:1 relative to the loudspeaker's weight. Always consult a qualified structural engineer when installing above an occupied area.

8.2 Aiming & Coverage

The DS-5 and DS-8 feature a $115^\circ \times 115^\circ$ symmetric dispersion, suitable for wide-area coverage at shorter throw distances. The DS-12 and DS-15 use a $90^\circ \times 60^\circ$ (H $\times$ V) pattern for controlled long-throw applications. Aim the speaker so that the on-axis direction points toward the centre of the listening area. For overlapping coverage zones, apply a 6 dB overlap rule: the -6 dB coverage edges of adjacent cabinets should meet at the boundary of each zone.

8.3 Wiring & Amplification

Use oxygen-free copper cable sized appropriately for cable run length to minimise resistive losses (target < 0.5 dB insertion loss). For runs above 30 m, use 2.5 mm^2 or larger conductor cross-section. Observe correct polarity (+ to +, – to –) to maintain phase coherence in multi-speaker arrays. Always match the amplifier's output impedance and power rating to the loudspeaker's nominal 8Ω impedance and rated power handling.

8.4 Controller Configuration

Louis Martin loudspeaker systems must be operated with an approved Louis Martin digital controller configured with the correct programme for the specific model. The controller provides protection filters, equalisation, and limiting tailored to each driver's characteristics. Operating without the correct controller voids the warranty and may result in driver damage.

8.5 Weatherproofing

Although rated IP54 (protected against dust ingress and water splashing from any direction), the loudspeaker terminals and cable entry points must be sealed with appropriate weatherproof glands or self-amalgamating tape in permanent outdoor installations. Inspect sealants annually and after extreme weather events.

8.6 Periodic Inspection

Inspect all mounting hardware, rigging points, and cable connections at a minimum of every six months. Look for signs of corrosion, mechanical wear, or cable abrasion. Replace any suspect hardware immediately. Keep a maintenance log for each installation.

9. Warranty Information

Louis Martin loudspeakers and electronics are covered against defects in workmanship or materials for a period of **two (2) years** from the original date of purchase. The warranty does not cover any item which, in Louis Martin's opinion, has failed due to user abuse, accidental damage, or unauthorised modifications.

Warranty Claim Procedure

1. Contact your authorised Louis Martin distributor or service centre.
2. Provide proof of purchase (original invoice or receipt) showing the purchase date.
3. Adequately pack the defective item to prevent transit damage.
4. Dispatch the item pre-paid (freight costs to the service centre are the owner's responsibility).
5. The item will be repaired or replaced at no charge for materials or labour if the fault is covered under warranty.

■ **Warranty is void if the system is operated without the correct Louis Martin digital controller and approved programme.**

10. Contact & Support

Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Warranty	Contact your local authorised Louis Martin distributor

For technical support, installation advice, or to locate your nearest authorised dealer, please visit our website or send an email to the address above. Our team of audio specialists is available to assist with system design, controller programming queries, and warranty claims.

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